

ANTS AND TERMITES

A



FOR EVERY HUMAN, there are 1,000,000 ants. Ants and termites are social insects that live in large colonies and have developed complex systems of communication. Ants are found worldwide, but, like termites, most of the 9,500 species of ant live in the tropics. There are more than 2,400 types of termite; many are blind, spending their lives inside nests, never seeing the light of day.

Ant nest

Most ants live in nests or colonies, usually underground. However, weaver ants build nests out of leaves in trees, and army ants build "live nests" of worker ants. Normally, there is one queen in a nest, but there are sometimes several. Nests of Australian bull ants contain up to 600 ants, while some wood ants' nests can house more than 300 million.

"Live nest" made by army ants

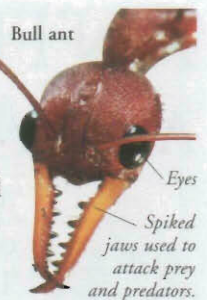


Ants

Ants have two pairs of compound eyes, three single eyes, or ocelli, two antennae, and three pairs of legs. Only queens and males have wings. A narrow waist connects the thorax and abdomen. Ants undergo complete metamorphosis, from an egg to larval and pupal stages, before emerging as adults. They live in huge groups and each ant has a particular role. The queen runs the nest and mates with male ants. Workers



Antennae are used to pick up the scent of pheromones.



Defence

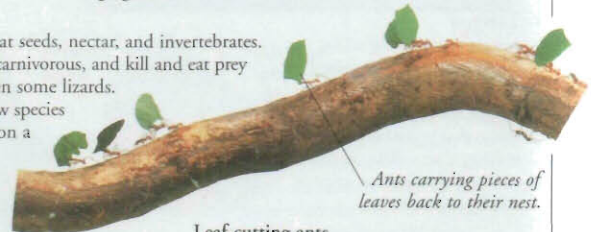
If a nest is attacked, the ants release pheromones to warn each other. Most run for cover, but soldier ants get aggressive and defend the colony. They attack enemies with their large jaws, or sting them and inject formic acid, which causes extreme pain. Some ants even explode to shower an attacker in venom.

Communication

Ants lay trails of pheromones – chemicals that smell – so that other ants can follow them by using their sensitive antennae to pick up the smell. This helps foraging teams home in on food.

Feeding

Many ants are omnivores and eat seeds, nectar, and invertebrates. Army and driver ants are more carnivorous, and kill and eat prey such as worms, spiders, and even some lizards. Leaf-cutting ants are one of a few species of herbivorous ants. They feed on a type of fungus, which grows on the chewed-up remains of leaves and flowers that the ants take back to their nests.



Termites

Although often called white ants, termites belong to a totally different order, the Isoptera. Like ants, termites live in large colonies. Unlike ants, termites do not have waists, and the male, called a king, does not die after mating, but lives with the queen. They do not go through complete metamorphosis, but grow up gradually through several nymphal stages.

Soldiers

Like ants, termites have soldiers. Termites cannot sting, but defend themselves in other ways. Some soldiers have large jaws that can cut through flesh; others squirt a poisonous sticky liquid from a special nozzle on their heads. Some nests have no soldiers – the termites defend themselves by vibrating their bodies against the side of their nest, making the sound of a hissing snake.



Queen and king

A queen termite can reach more than 15 cm (6 in) in length. Her ovaries make her so large. She can lay up to 30,000 eggs a day. The king remains by the queen's side and mates with her several times to fertilize all the eggs.

Queen



Fungus gardens are areas where fungi grow on termites' faeces and break down the cellulose within them. The termites feed on the products released and the fungi itself.



"Chimneys" allow warm air to rise and escape.

Solid outer walls are up to 50 cm (20 in) thick.

Air channel

Ground level

Nurseries

Royal chamber

Thick pillar support nest.

Workers

Worker termites build the nest, collect food, and feed the soldiers, king, and queen. The nest is made from saliva, soil, and their own faeces. Most workers feed on wood and have microscopic organisms in their guts to break down the wood into a more easily digested form.



WOOD ANT

SCIENTIFIC NAME	<i>Formica rufa</i>
ORDER	Hymenoptera
FAMILY	Formicidae
DISTRIBUTION	Europe
HABITAT	Woods and forests
DIET	Omnivorous, feeding on seeds and invertebrates
SIZE	Workers 6–8 mm (0.24–0.31 in) in length; queen 10–13 mm (0.4–0.5 in) in length
LIFESPAN	Workers live for 3–4 months; the queen lives for about 15 years

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